# Effect of a Daily Supplement of Soy Protein on Body Composition and Insulin Secretion in Postmenopausal Women

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Objective: To determine if a supplement of soy protein improves body composition, body fat distribution, and glucose and insulin metabolism in non-diabetic postmenopausal women compared to an isocaloric casein placebo. Design: Randomized, double-blind, placebo-controlled 3-month trial Setting: Clinical Research Center Patients: 15 postmenopausal women Interventions: CT scans at L4/L5, dual energy x-ray absorptiometry (DXA), hyperglycemic clamps Main outcome measures: Total fat, total abdominal fat, visceral fat, subcutaneous abdominal fat, and insulin secretion. Results: Weight by DXA did not change between groups (+1.38 ± 2.02 kg for placebo vs. +0.756 ± 1.32 kg for soy, p=0.48, means ± S.D.). Total and subcutaneous abdominal fat increased more in the placebo compared to the soy group (for differences between groups in total abdominal fat: +38.62 ± 22.84 cm2 for placebo vs. –11.86 ± 31.48 cm2 for soy, p=0.005; subcutaneous abdominal fat: +22.91 ± 28.58 cm2 for placebo vs. –14.73 ± 22.26 cm2 for soy, p=0.013). Insulin secretion, visceral fat, total body fat, and lean mass did not differ between groups. Isoflavone levels increased more in the soy group. Conclusion: A daily supplement of soy protein prevents the increase in subcutaneous and total abdominal fat observed with an isocaloric casein placebo in postmenopausal women.